INFOMDW: Data Wrangling

Data Reduction

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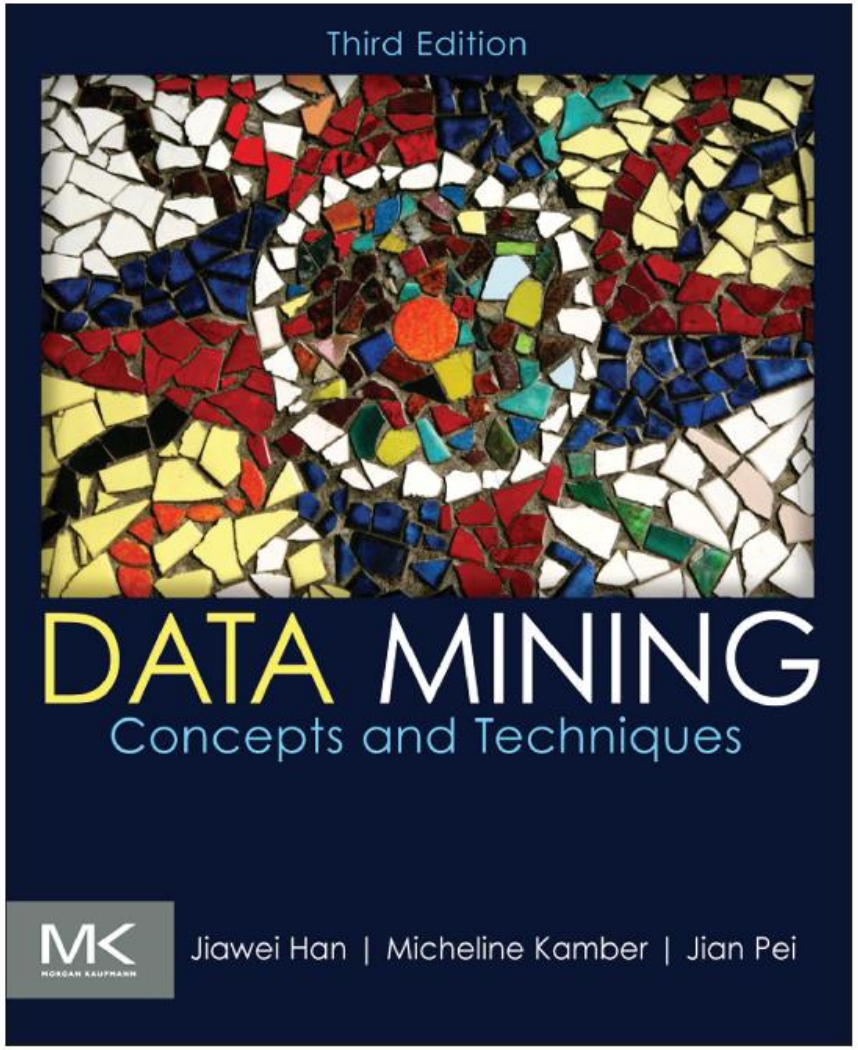
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Reading Material for Today

- Data Mining: Concepts and Techniques Book

  CH 3.4

- Other research papers mentioned in the slides



Topics for Today

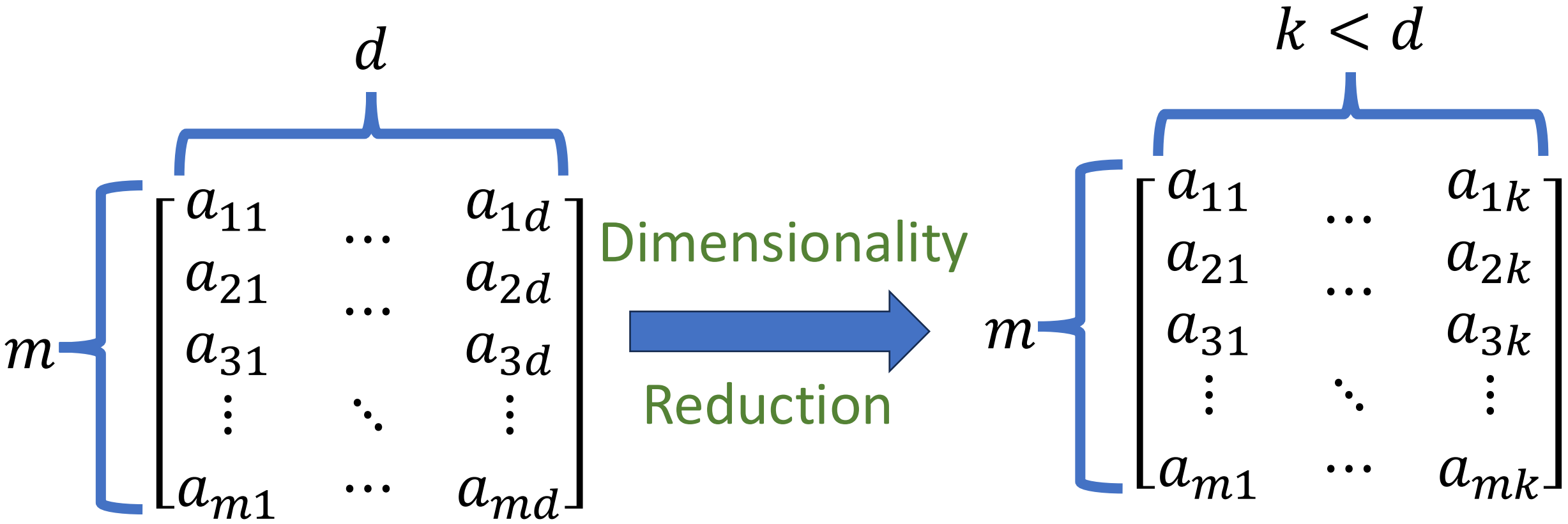
- Data reduction
  - Reducing the number of features
    - Dimensionality reduction
    - Manifold learning
  - Reducing the number of tuples

Data Reduction

- **Data reduction**: obtain a reduced representation of the dataset
  - Much smaller in volume but yet produces the same (or almost the same) analytical results
- Why data reduction?
  - A dataset could be extremely large – Complex data analysis may take a very long time to run on the complete dataset.

Dimensionality Reduction

Dimensionality reduction

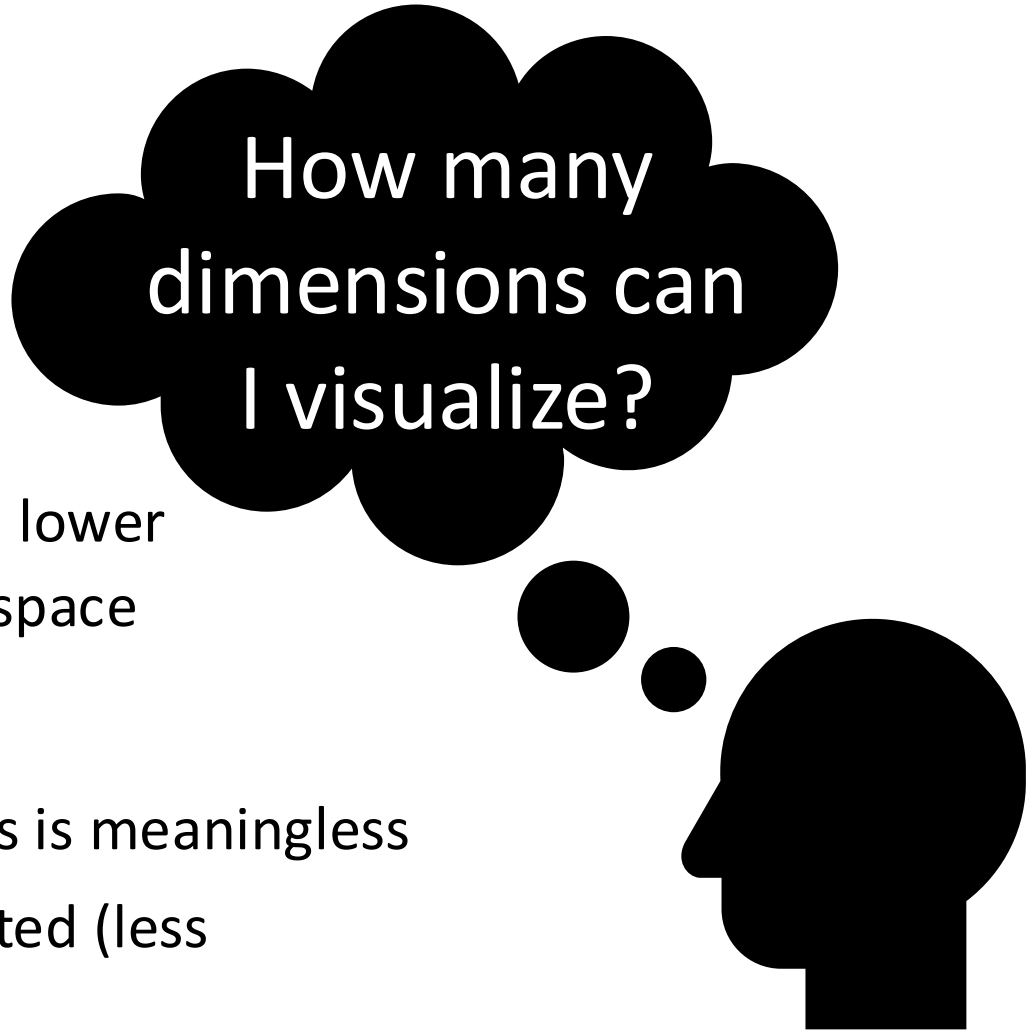


Dimensionality Reduction

- Curse of dimensionality
  - When dimensionality increases, data becomes increasingly sparse
  - Density and distance between points, which is critical to clustering, outlier analysis, becomes less meaningful
  - The possible combinations of subspaces will grow exponentially
- Dimensionality reduction
  - Avoid the curse of dimensionality
  - Help eliminate irrelevant features and reduce noise
  - Reduce time and space required in data mining
  - Allow easier visualization

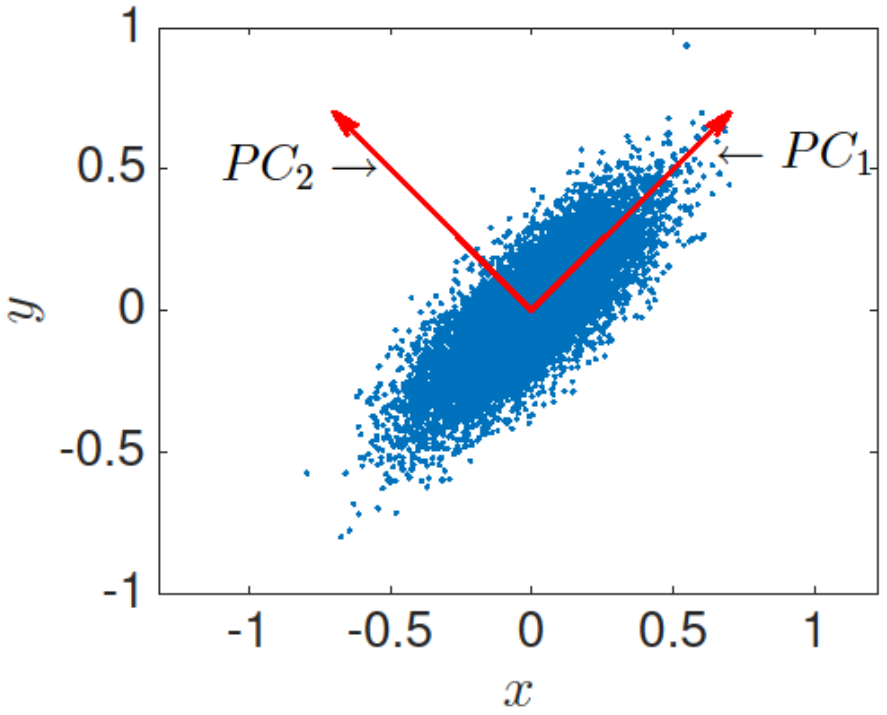
Dimensionality reduction

- Applications
  - Visualization – Exploration – Compression
- Key observation
  - data points all lie close to a manifold of much lower dimensionality than that of the original data space
- Key considerations
  - Dimensionality reduction by deleting columns is meaningless
  - The columns in the new matrix are uncorrelated (less redundancy)
  - Columns might become less interpretable

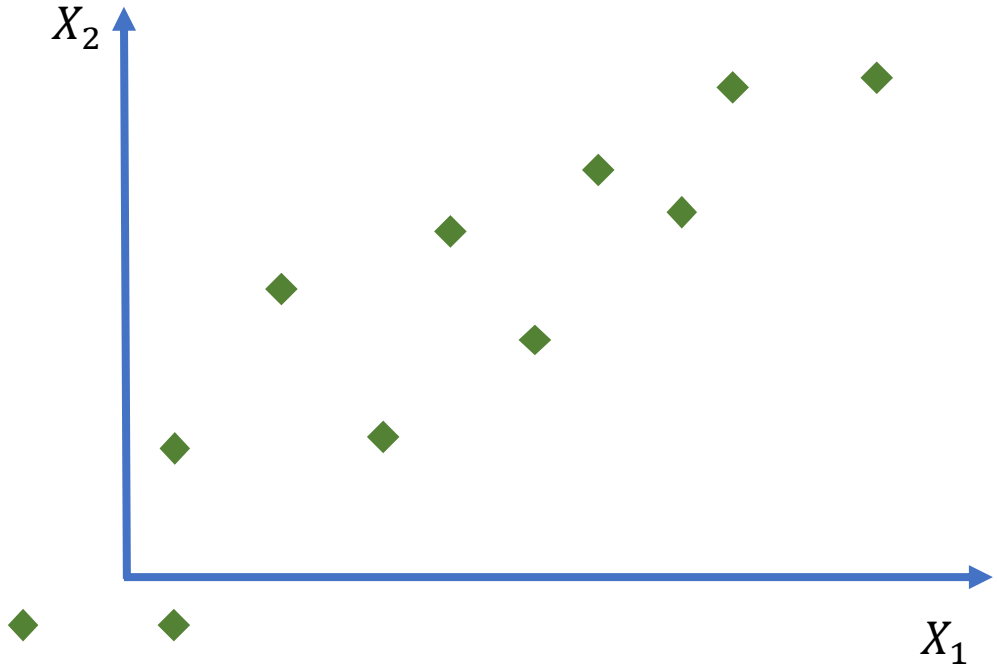


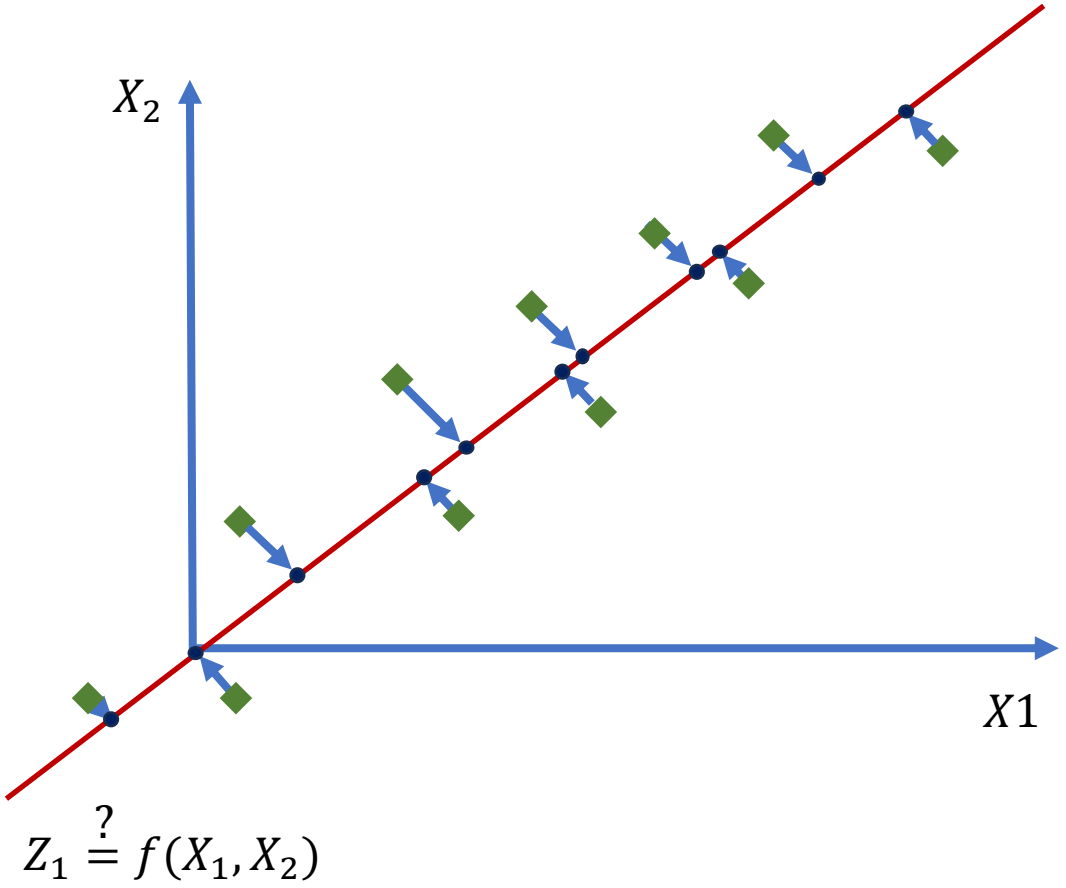
Principal Component Analysis (PCA)

- Find a projection that captures the largest amount of variation in data
- The original data are projected onto a much smaller space, resulting in dimensionality reduction.
- We find the eigenvectors of the covariance matrix, and these eigenvectors define the new space



Principal Component Analysis (PCA)





$Z_1 \stackrel{?}{=} f(X_1, X_2)$

Minimize reconstruction error

$$X_{m\times d} \xrightarrow{PCA} Z_{m\times k} \qquad k \ll d$$

$$Z_{m\times k} = X_{m\times d} U_{d\times k} \qquad U^T U = I$$

- Find $U_{d\times k}$ that minimizes the difference

$$\|Z_{m\times k}(U_{d\times k})^T - X_{m\times d}\|^2$$

Maximize variance

$$X_{m\times d} \xrightarrow{PCA} Z_{m\times k} \qquad k \ll d$$

$$Z_{m\times k} = X_{m\times d} U_{d\times k} \qquad U^T U = I$$

- Find $U_{d\times k}$ where

$u_1^T \Sigma u_1 = \lambda_1$ $\quad$ u_1 is the eigenvector with the largest eigenvalue λ_1

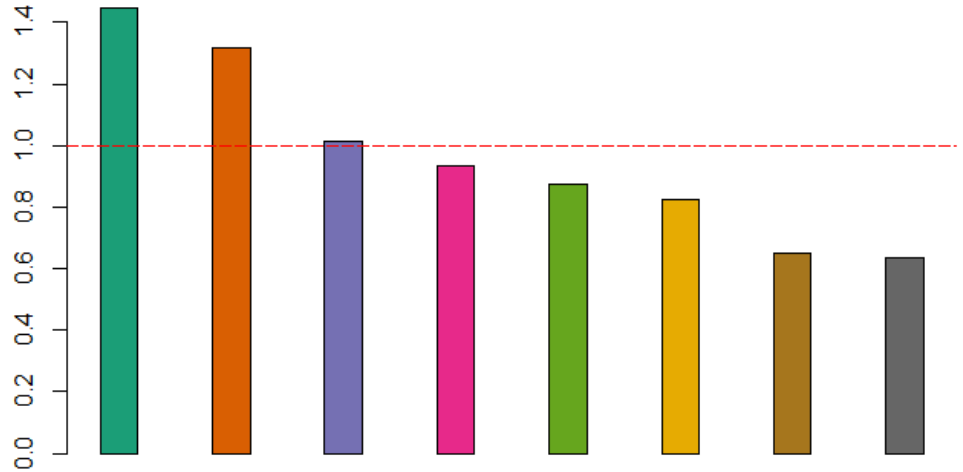
$u_2^T \Sigma u_2 = \lambda_2$ $\quad$ u_2 is the eigenvector with the 2nd largest eigenvalue λ_2

Principal Component Analysis (Steps)

- Given: N data vectors from d-dimensions, find $k \leq d$ principal components that can accurately represent the data
  - Normalize input data: each attribute falls within the same range
  - Compute k orthonormal (unit) vectors, i.e., principal components
  - Each input data (vector) is a linear combination of the k principal component vectors
  - The principal components are sorted in order of decreasing "significance" or strength
  - The components are sorted
    - Reduce the data dimensionality by eliminating the weak components
    - Weak components have low variance
    - We can reconstruct a good approximation of the original data using strong components
- Works for numeric data only

PCA for dimensionality reduction

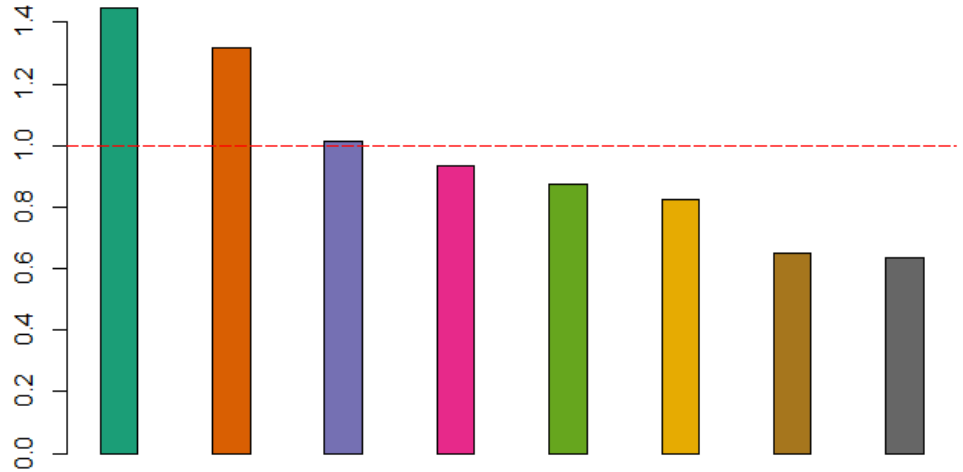
- Can ignore the components of lesser significance.



- You lose some information, but if the eigenvalues are small, it is not much
  - d dimensions in original data
  - calculate d eigenvectors and eigenvalues
  - choose first k eigenvectors – final dataset has only k dimensions

PCA for dimensionality reduction

- Can ignore the components of lesser significance.



- You lose some information: minimize the information loss
  - Kaiser criterion: One common heuristic is to
    - keep components with eigenvalues greater than 1
  - Better to choose k based on cumulative explained variance (80% or more)

Singular value decomposition (SVD)

- Historical overview
  - Eugenio Beltrami (1873): proved the existence of SVD
  - Camille Jordan (1874): theory of singular values for bilinear forms
  - Gene Golub and W. Kahan (1965): practical algorithm for computing the SVD

Singular value decomposition (SVD)

- Decompose a matrix $X_{m\times n}$ into 3 matrices
  - $X_{m\times n} = U\Sigma V^T$
  - $U_{m\times m}, V^t_{n\times n}$ are orthogonal matrices and $\Sigma_{m\times n}$ is a diagonal matrix
  - When $m = n$:

$$\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ a_{21} & \cdots & a_{2n} \\ a_{31} & & a_{3n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \xrightarrow{\text{SVD}} \begin{bmatrix} \cdots & \cdots & \cdots \\ u_1 & \cdots & u_n \\ \vdots & \ddots & \vdots \\ \cdots & \cdots & \cdots \end{bmatrix} \begin{bmatrix} \sigma_1 & \cdots & 0 \\ 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_n \end{bmatrix} \begin{bmatrix} \cdots & v_1^T & \cdots \\ \cdots & v_2^T & \cdots \\ \vdots & \vdots & \vdots \\ \cdots & v_n^T & \cdots \end{bmatrix}$$

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n$$

Singular value decomposition (SVD)

- $m > n$

$$\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ a_{21} & \cdots & a_{2n} \\ a_{31} & & a_{3n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}$$

SVD

$$\underbrace{\begin{bmatrix} \cdots & \cdots & \cdots & \cdots & \cdots \\ u_1 & \ddots & u_n & u_{n+1} & u_m \\ \cdots & \cdots & \cdots & \cdots & \cdots \end{bmatrix}}_{m \times m} \underbrace{\begin{bmatrix} \sigma_1 & \cdots & 0 \\ 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_n \\ \cdots & \cdots & \cdots \\ 0 & 0 & 0 \end{bmatrix}}_{m \times n} \underbrace{\begin{bmatrix} \cdots & v_1^T & \cdots \\ \cdots & v_2^T & \cdots \\ \vdots & \vdots & \vdots \\ \cdots & v_n^T & \cdots \end{bmatrix}}_{n \times n}$$

Singular value decomposition (SVD)

- $n > m$

$$\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ a_{21} & \cdots & a_{2n} \\ a_{31} & & a_{3n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}$$

SVD

$$\begin{bmatrix} \cdots & \cdots & \cdots \\ u_1 & \cdots & u_m \\ \vdots & \ddots & \vdots \\ \cdots & \cdots & \cdots \end{bmatrix} \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \sigma_m \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \cdots & v_1^T & \cdots \\ \cdots & v_2^T & \cdots \\ \vdots & \vdots & \vdots \\ \cdots & v_m^T & \cdots \\ \vdots & \vdots & \vdots \\ \cdots & v_n^T & \cdots \end{bmatrix}$$

$m \times m$ $\quad$ $m \times n$ $\quad$ $n \times n$

How do we compute SVD?

- Find the n eigencevtors v_i and eigenvalues λ_i of AA^T
  - Create a matrix V from the normalized eigenvectors $\begin{bmatrix} \cdots & v_1^T & \cdots \\ \cdots & v_2^T & \cdots \\ \vdots & \vdots & \vdots \\ \cdots & v_n^T & \cdots \end{bmatrix}$
  - Create a diagonal matrix Σ from the square roots of the eigenvalues

$$\Sigma = \begin{bmatrix} \sigma_1 & \cdots & 0 \\ 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_n \end{bmatrix}, \quad \sigma_i = \sqrt{\lambda_i} \quad \text{and} \quad \sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n$$

  - Compute U: $A = U\Sigma V^T \implies U\Sigma = AV \implies U = AV\Sigma^{-1}$

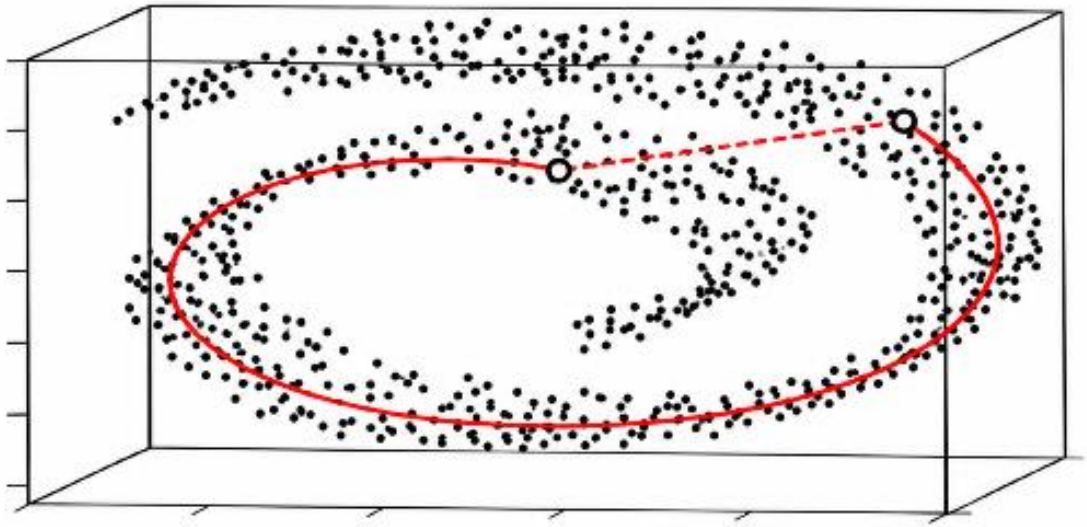
Attribute Subset Selection for Data Reduction

- Another way to reduce dimensionality of data
- Redundant attributes
  - Duplicate much or all of the information contained in one or more other attributes
  - E.g., purchase price of a product and the amount of sales tax paid
- Irrelevant attributes
  - Contain no information that is useful for the data mining task at hand
  - E.g., students' ID is often irrelevant to the task of predicting students' GPA

Manifold Learning

Geodesic Distance

To preserve the shape, preserve the geodesic distance

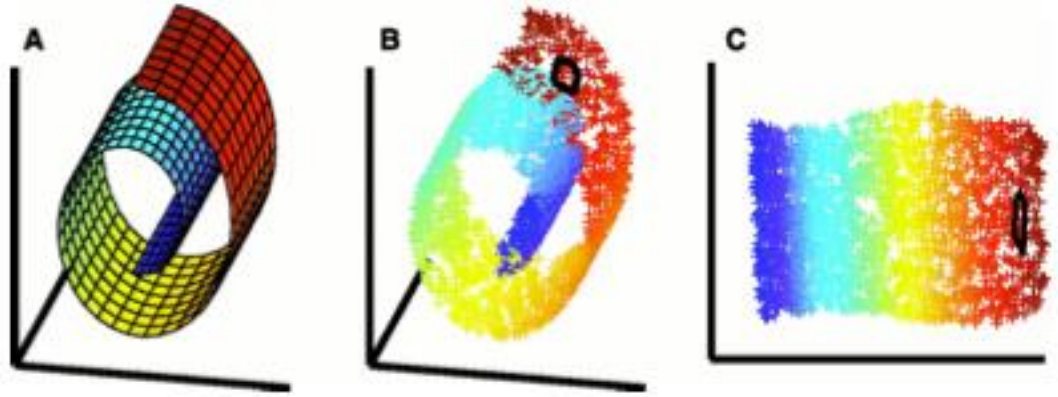






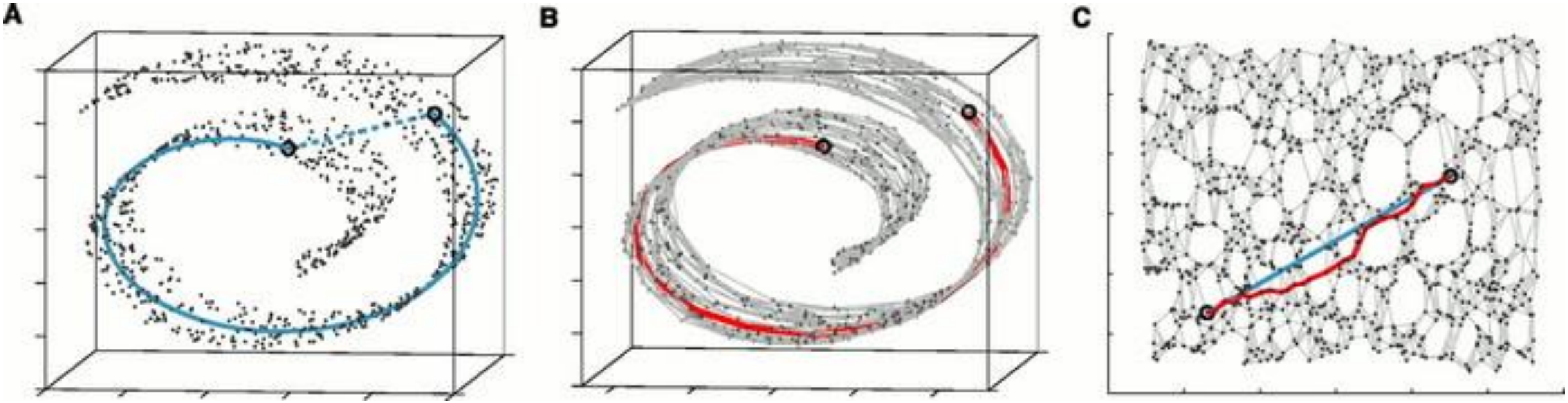
ISOMap

- Estimate the geodesic distance between faraway points.
- For neighboring points Euclidean distance is a good approximation to the geodesic distance.
- For faraway points estimate the distance by a series of short hops between neighboring points.
  - Find shortest paths in a graph with edges connecting neighboring data points

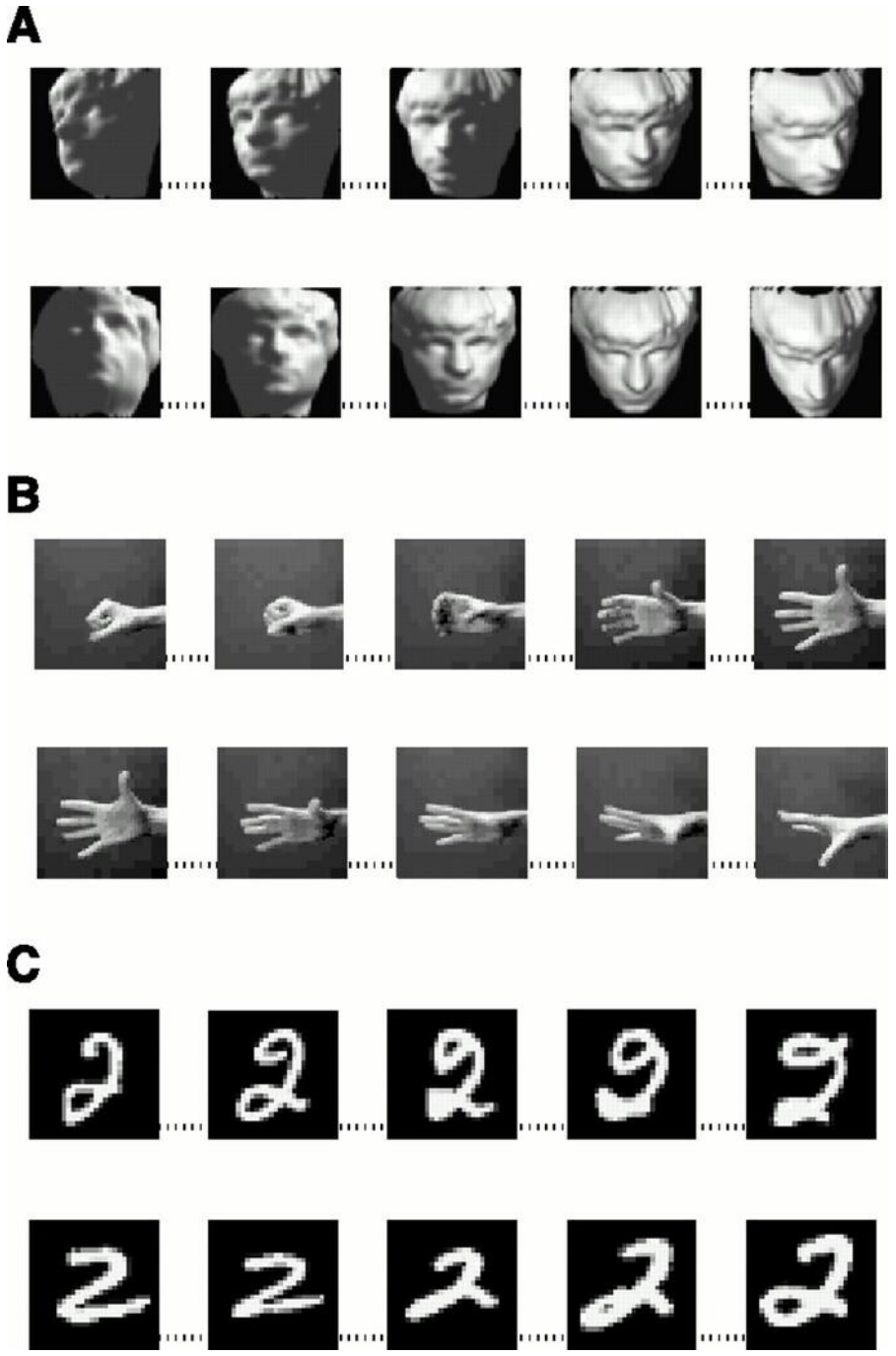


ISOMAP Joshua B. Tenenbaum, Vin de Silva, and John C. Langford [Science 2000]
https://wearables.cc.gatech.edu/paper_of_week/isomap.pdf

ISOMap

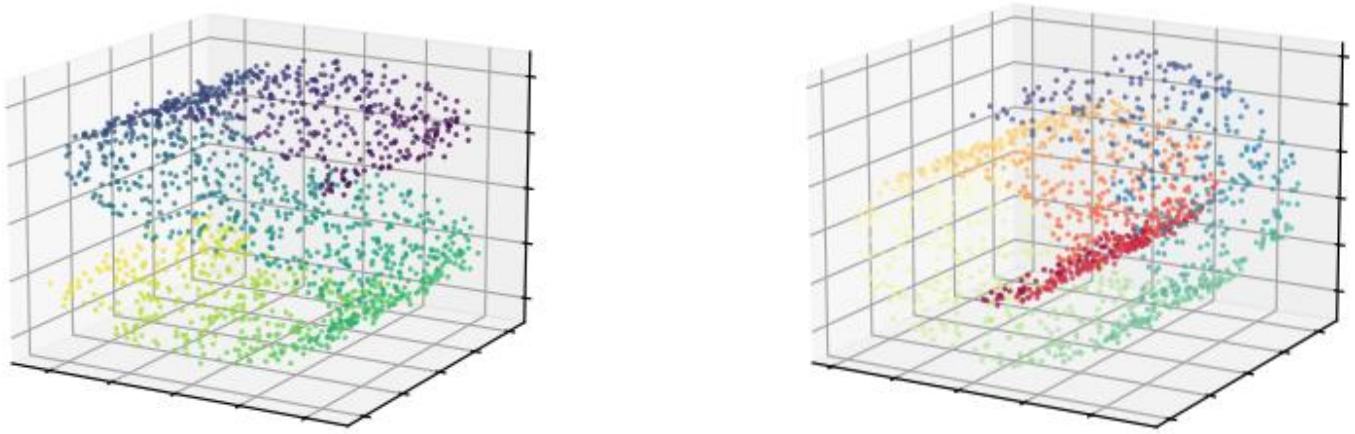


ISOMap



Locally Linear Embedding (LLE)

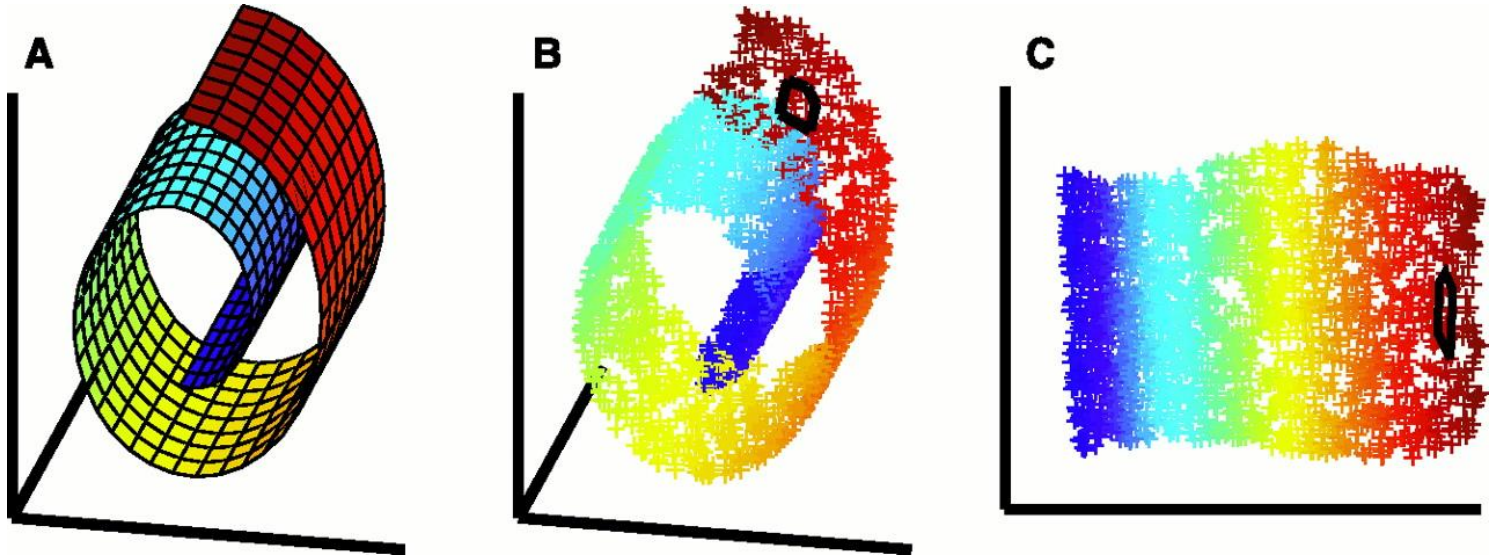
- Intuition: data points all lie close to a manifold of much lower dimensionality than that of the original data space
- Examples: S-Shape, Swiss-roll



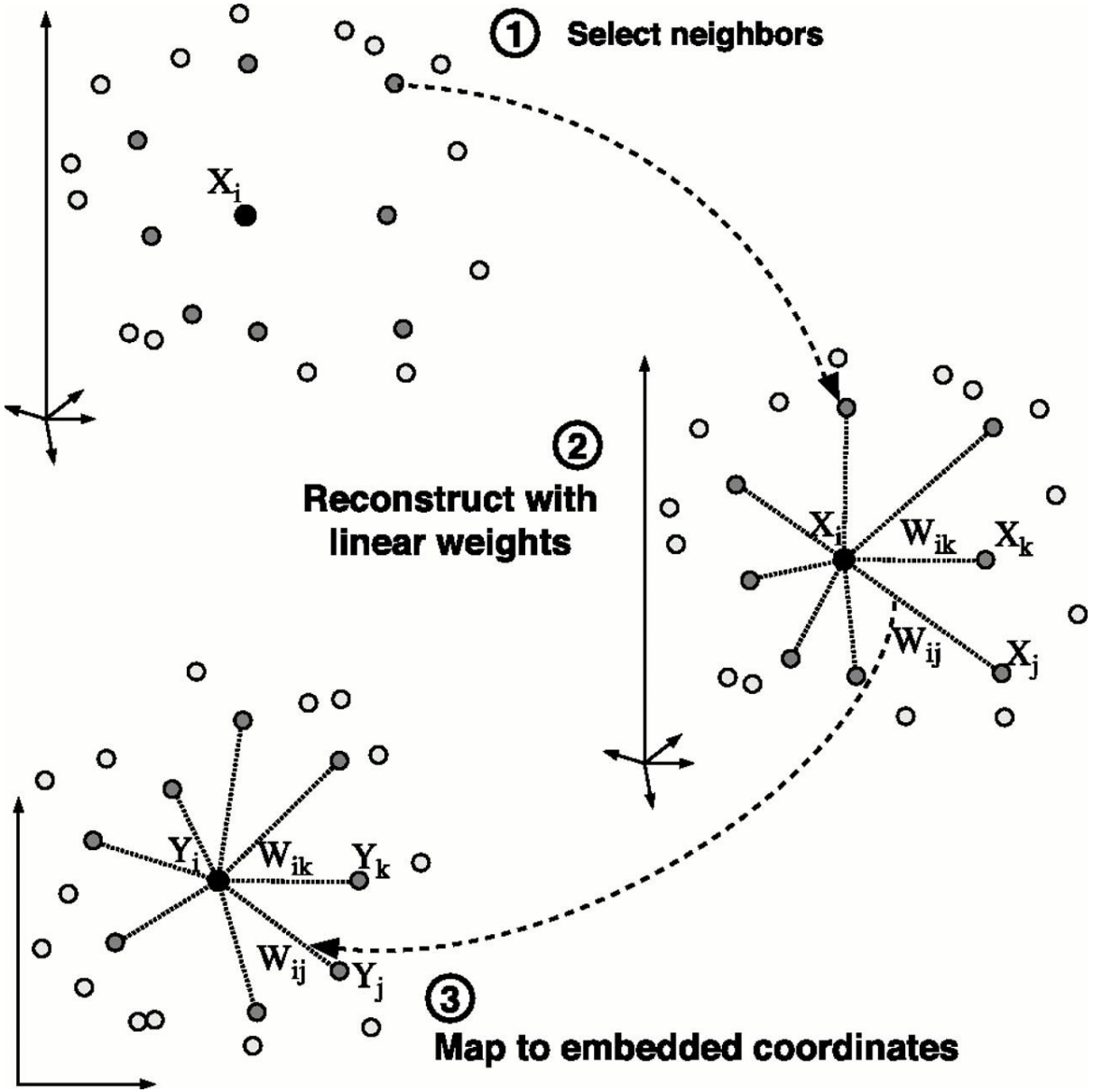
LLE, Sam T. Roweis1 and Lawrence K. Saul [Science 2000]
https://wearables.cc.gatech.edu/paper_of_week/isomap.pdf

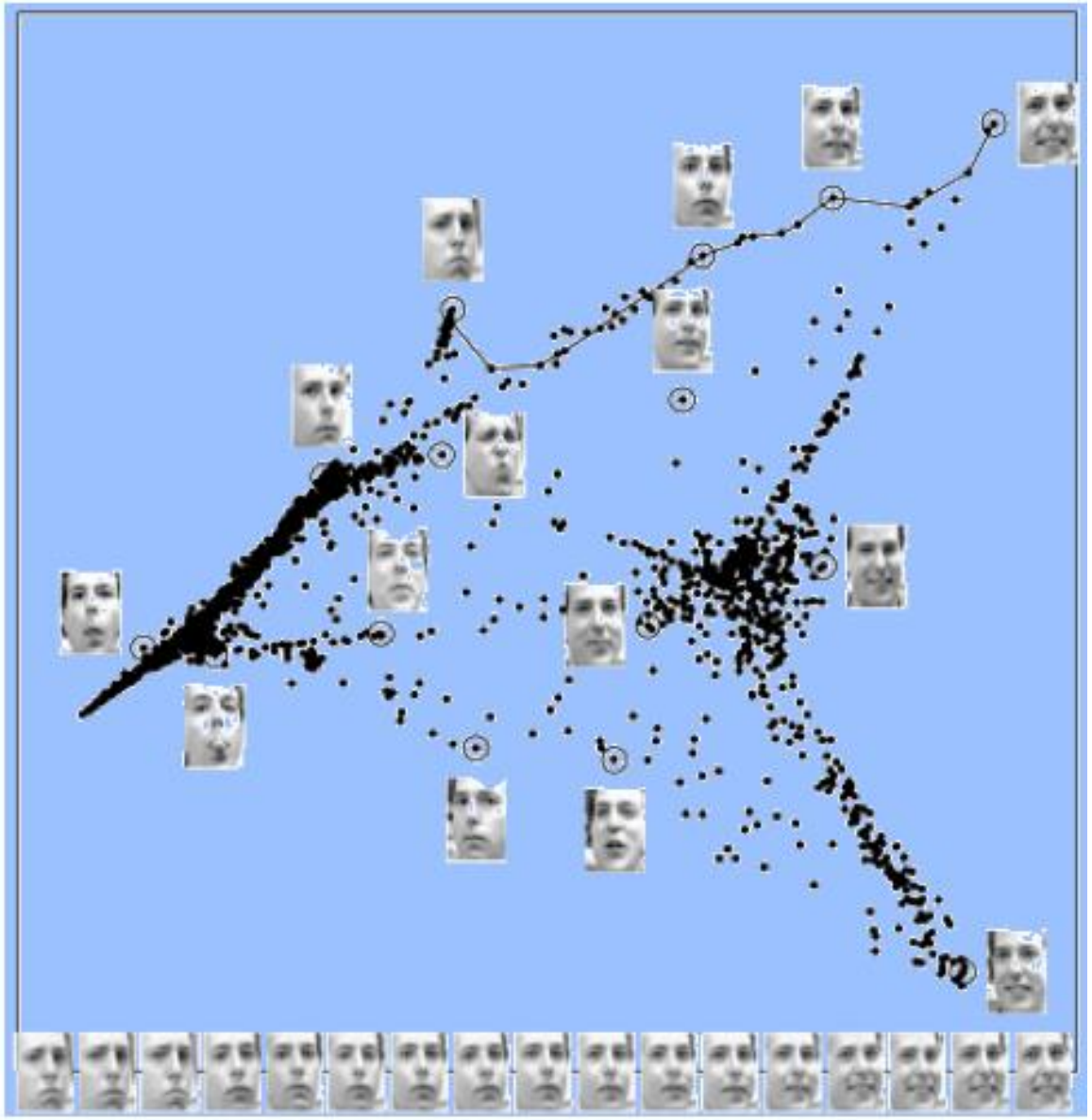
LLE

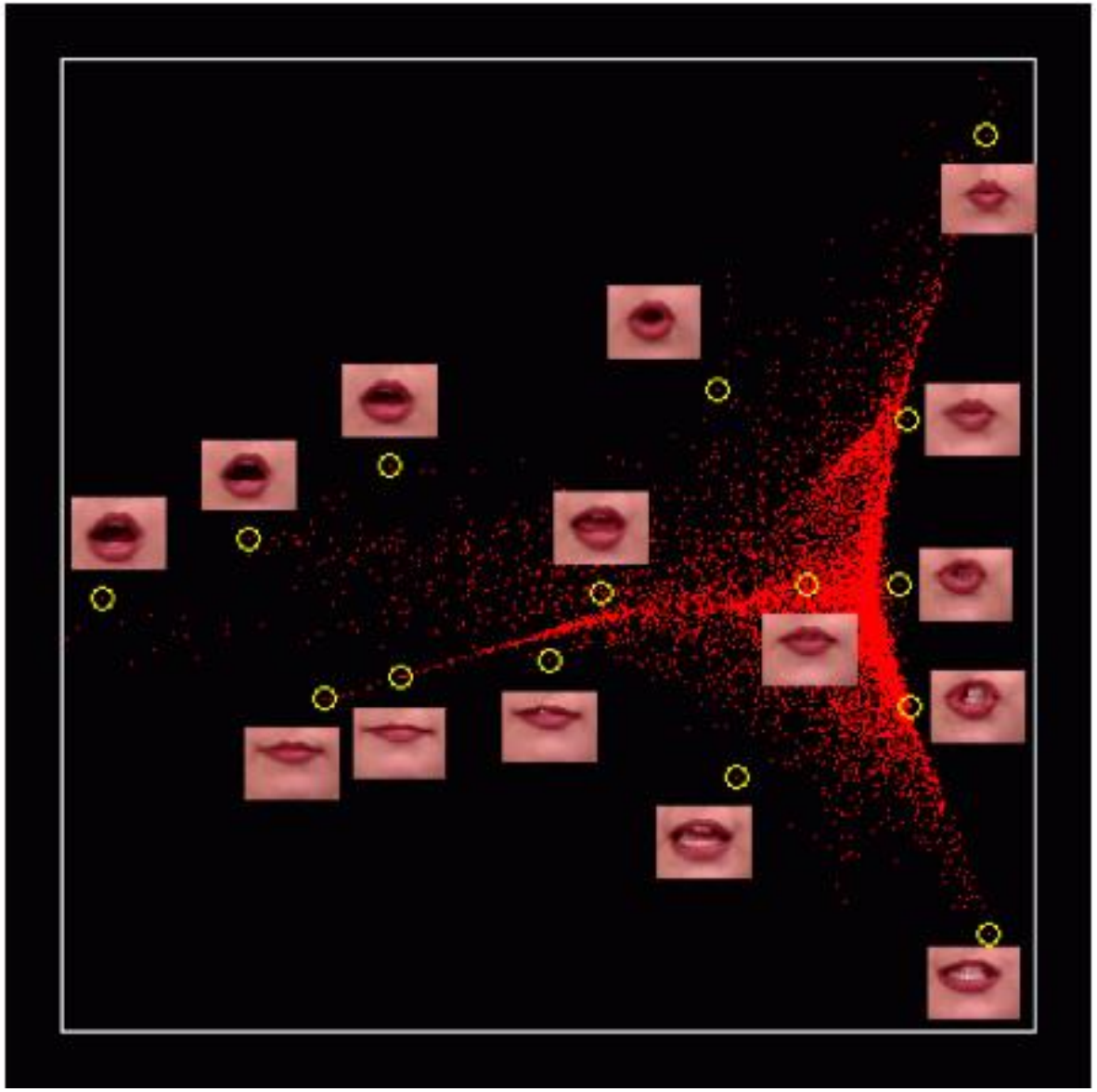
- Each data point and its neighbors lie on or close to a locally linear patch of the manifold



LLE


① Select neighbors
X_i
② Reconstruct with linear weights
X_i
W_{ik}
X_k
W_{ij}
X_j
Y_i
W_{ik}
Y_k
W_{ij}
Y_j
③ Map to embedded coordinates





More Techniques

- You may also look at
  - Spectral embedding
  - Multi-dimensional Scaling (MDS)
  - t-distributed Stochastic Neighbor Embedding (t-SNE)
  - Uniform Manifold Approximation and Projection (UMAP)

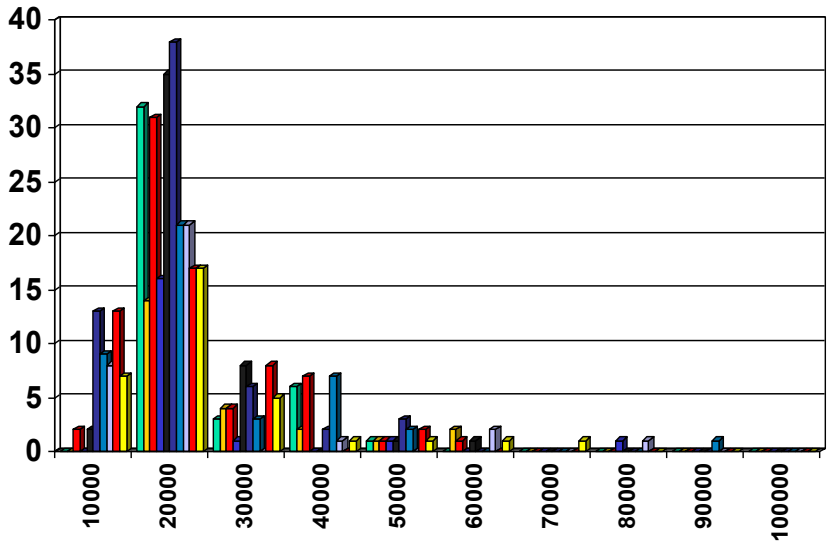
Numerosity reduction

Data Reduction (instances or examples level)

- Linear regression
  - Data modeled to fit a straight line
  - Often uses the least-square method to fit the line
- Multiple regression
  - Allows a response variable Y to be modeled as a linear function of multidimensional feature vector
- Log-linear model
  - Approximate data by a function whose logarithm is linear

Histograms for Instance Level Data Reduction

- Divide data into buckets and store average (sum) for each bucket
- Partitioning rules:
  - Equal-width: equal bucket range
  - Equal-frequency (or equal-depth)



Clustering-Based Data Reduction

- Partition data set into clusters based on similarity, and store cluster representation (e.g., centroid and diameter) only
- Can be very effective if data is forms compact, well-separated groups
- Less effective for diffusely distributed data
- There are many choices of clustering definitions and clustering algorithms

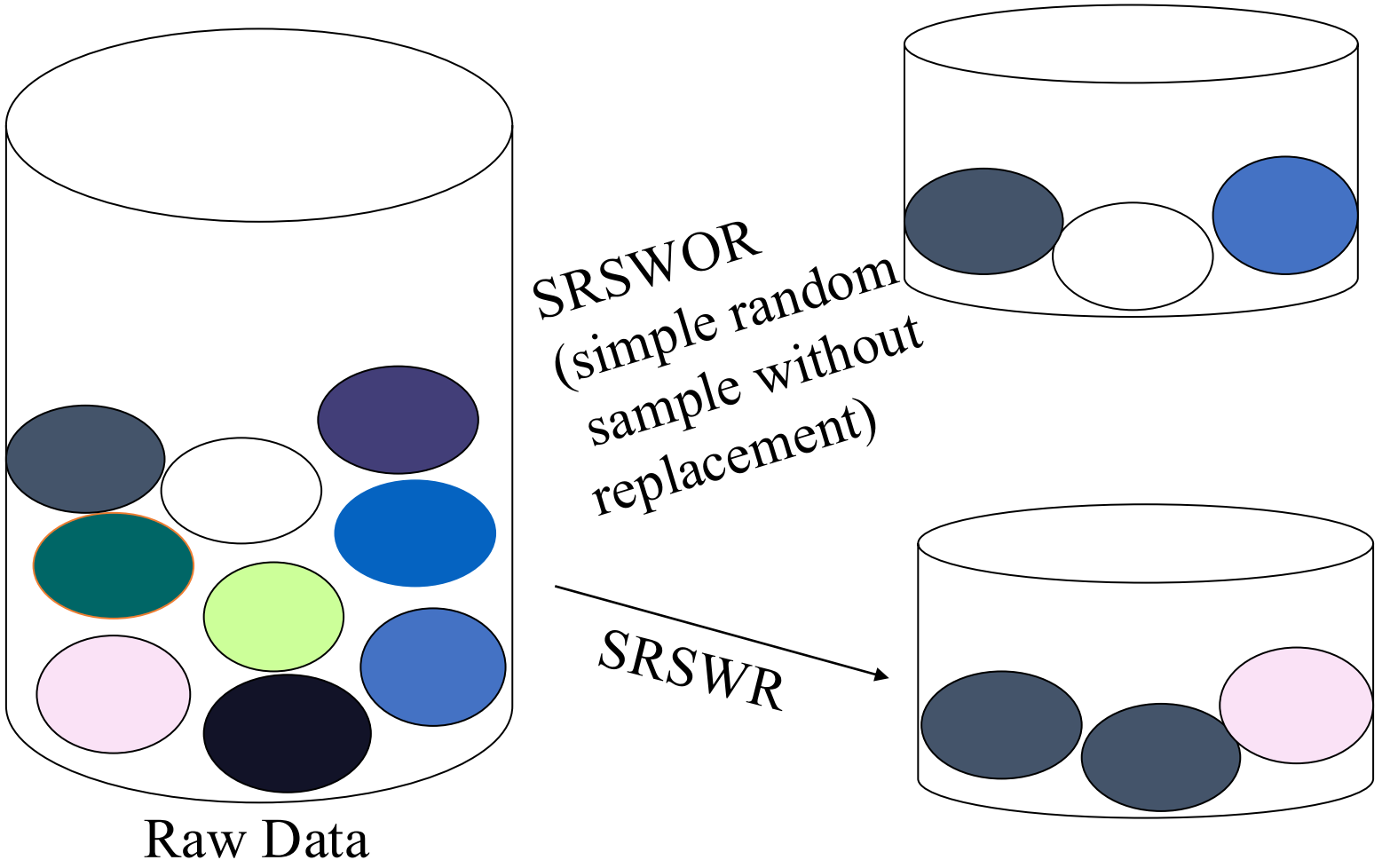
Sampling-Based Data Reduction

- Sampling: obtaining a small sample s to represent the whole data set N
- Allow a mining algorithm to run in complexity that is potentially sub-linear to the size of the data
- Key principle: Choose a representative subset of the data
  - Simple random sampling may have very poor performance in the presence of skew
  - Develop adaptive sampling methods, e.g., stratified sampling
- Note: Sampling may not reduce database I/Os (page at a time)

Types of Sampling

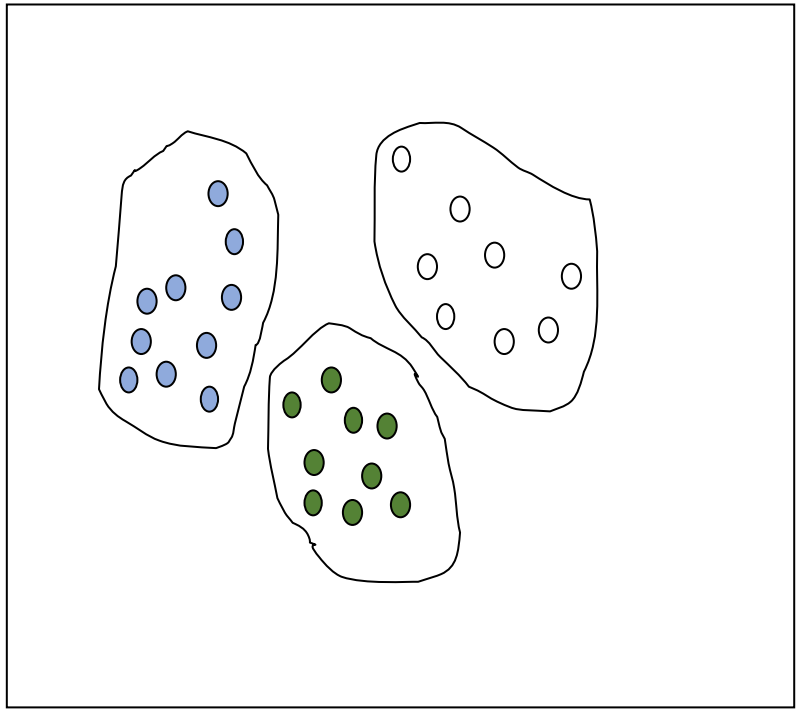
- Simple random sampling
  - There is an equal probability of selecting any particular item
- Sampling without replacement
  - Once an object is selected, it is removed from the population
- Sampling with replacement
  - A selected object is not removed from the population
- Stratified sampling:
  - Partition the data set, and draw samples from each partition (proportionally, i.e., approximately the same percentage of the data)
  - Used in conjunction with skewed data

Sampling – with or without Replacement

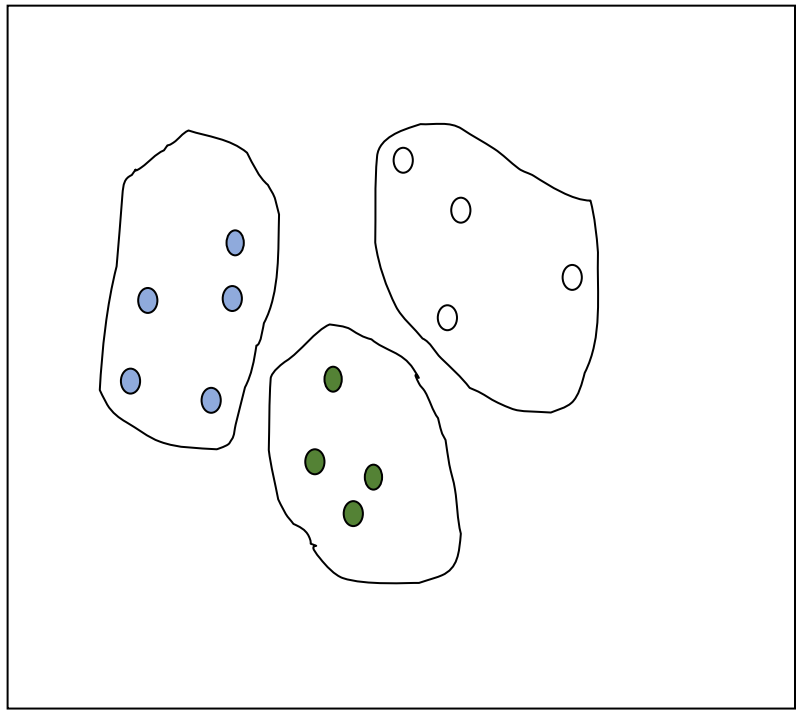


Sampling – Cluster or Stratified Sampling

Raw Data



Cluster/Stratified Sample



Concept Hierarchy Generation

- Some hierarchies can be automatically generated based on the analysis of the number of distinct values per attribute in the data set
  - The attribute with the most distinct values is placed at the lowest level of the hierarchy



Wrap-Up

Summarize what you learned today in 2-minutes



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